

■ Chick Hatching Science!

Ms. Wotkyns' Classroom Hatching Guide
Kittridge Elementary STEM School • Van Nuys, CA

We're going to hatch real chicken eggs — and YOU are the scientist! ■
It takes about **21 days** for a chick to hatch. Follow these steps carefully!

■ Step 1: Get the Incubator Ready

- Set up the incubator **1–2 days before** the eggs go in — like warming up an oven!
- Target temperature: **99–102°F** (that's warm, like a cozy nest!)
- Target humidity: **45–55%** — check with a digital hygrometer!
- Make sure there's **good airflow** — baby chicks need fresh air to grow!
- Clean and sanitize the incubator before adding eggs.
- *Science Tip: Run the empty incubator for a whole day to make sure the temperature stays steady!*

■ Step 2: Choosing and Setting the Eggs

- **Do NOT wash the eggs!** They have a natural coating called a "bloom" that protects them.
- Store eggs **pointy-end down** for 1–2 days before placing them in the incubator.
- Use a pencil to draw an **X on one side** and an **O on the other side** — this helps you track turning!

■ Step 3: Days 1–18: Incubation

■ Temperature & Humidity

- Keep temperature at **99–102°F** and humidity at **45–55%**.
- Check levels **at least twice a day!**

■ Turning the Eggs

- Turn eggs **at least 3 times per day** (always an odd number of turns).
- An automatic egg turner does this for you — like a robot helper!
- If turning by hand: **wash your hands first** and be very gentle.
- **Stop turning at Day 18** — that's when lockdown begins!

■ Candling (Optional but Super Cool!)

- Shine a bright light through the egg in a dark room to see inside!
 - **Day 7:** Look for a dark spot with tiny veins — that's a live embryo! ■■
 - **Day 14:** The embryo fills most of the egg — you can see the air cell!
- Remove any **clear (unfertilized)** eggs or ones with a blood ring.
- *Science Tip: Candle in a dark room for the best view. A flashlight works great!*

■ Step 4: Day 18: LOCKDOWN!

This is a critical phase — handle with care! ■

- **STOP turning eggs** — lay them on their sides in hatching trays.
- **Increase humidity to 65–70%** — this stops chicks from getting stuck in their shell!
- **Do NOT open the incubator** during lockdown unless absolutely necessary.
- Add water through the outside ports if you can, to avoid opening the lid.
- **Keep checking that temperature is still on target!**

■ Step 5: Days 19–21: Hatching Time!

- Around **Day 19–20**, look for a **pip** — the chick's first crack in the shell!
 - **External pip:** A tiny hole or crack on the outside of the shell ■
 - **Internal pip:** Chick breaks into the air cell — you might hear chirping! ■
- Once pipping starts, it can take **12–24+ hours** for the chick to fully hatch. **Be patient!**
- **Do NOT help the chick hatch** unless it has been stuck for 24+ hours and is in distress.
- **Do NOT open the incubator** while chicks are hatching — a humidity drop can hurt unhatched eggs.
- Leave newly hatched chicks inside for **12–24 hours** to dry off and fluff up!

■ Step 6: After the Hatch!

- Move fluffy, dry chicks to a **brooder** (a warm box) set to **95°F** for the first week.
- Give them **chick starter feed** (crumbles) and **fresh water** right away!
- Lower the brooder temperature by **5°F each week** until they grow feathers (~6 weeks).
- Chicks are born with enough energy to go **2–3 days** without food — no rush!
- Clean and sanitize the incubator after the hatch is complete.

■ Scientist Tips for a Great Hatch! ■

- ■ Use a **digital thermometer AND hygrometer** — analog ones aren't accurate enough.
- ■ Keep the incubator **away from sunlight, drafts, and vents**.
- **Keep a hatch log!** Write down temperature, humidity, and what you observe every day.
- **Humidity matters A LOT** — don't ignore it!
- **Don't overcrowd** the incubator — eggs need space and airflow.
- **Fresh farm eggs** hatch at 80–90%. **Shipped eggs** may be lower (50–70%) — that's normal science!

■ Quick Reference Chart

Phase	Days	Temperature	Humidity	Key Task
■ Incubation	1–18	99–102°F	45–55%	Turn eggs 3x/day
■ Lockdown	18	99–102°F	65–70%	Stop turning!
■ Hatching	19–21	99–102°F	65–70%	Be patient!
■ Brooder	After hatch	Start 95°F	—	Feed & water

■ *Happy Hatching, Scientists!* ■

Guide adapted for Ms. Wotkyns' Class • Kittridge Elementary STEM School • Van Nuys, CA

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